

## Ex2) Scissor Jack

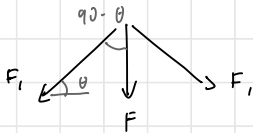
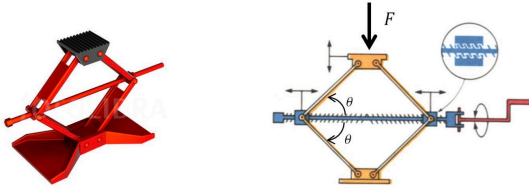
- Compute the torque at the handle to lift up the load given

- single-start, square thread,  $M10 \times 1.25$  with  $d_p = 9.1881$
- the lead screw friction  $\mu = 0.15$ ,
- the weight  $F = 4 \text{ kN}$  and  $\theta = 30^\circ$

$$\theta = 30^\circ, \quad d_p = 9.1881 \text{ mm}$$

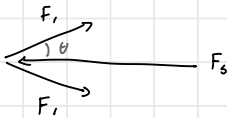
$$L = 1.25 \text{ mm}$$

$$\tan \lambda = \frac{L}{\pi d_p} = 0.0433$$



$$F = 2F_1 \cos(90^\circ - \theta)$$

$$\Rightarrow F_1 = \frac{F}{2 \sin \theta}$$



$$F_2 = 2F_1 \cos \theta = 2 \frac{F}{2 \sin \theta}$$

$$= \frac{F}{\tan \theta}$$

$$= \frac{4000}{\tan 30^\circ} = 6928.2 \text{ N}$$

Since there are 2 connected threads,

$$T_{\text{up}} = \cancel{2} \cdot \frac{F_2 d_p}{\cancel{2}} \cdot \frac{\mu + \tan \lambda}{1 - \mu \tan \lambda}$$

$$= 6928.2 \cdot 0.0091881 \cdot \frac{0.15 + 0.0433}{1 - 0.15 \cdot 0.0433} = 12.385 \text{ Nm}$$

$$T_{\text{down}} = 6928.2 \cdot 0.0091881 \cdot \frac{0.15 - 0.0433}{1 + 0.15 \cdot 0.0433} = 6.74 \text{ Nm}$$